

Rotating Group Assignment with a Guaranteed Companion: Coercion, Capacity, Leakage, and a Hybrid Local-Search / Constraint-Programming Solver

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Abstract

Institutions that periodically reassign people to small groups (lunch tables, seminar sections, project teams) want mixing without isolation: everyone should meet many others, and nobody should arrive among strangers. We formalise the *rotating group assignment problem with an anchor guarantee*: over a sequence of periods, partition a population into capacity-constrained groups, stratified into blocks in some periods, so that every participant who submitted a list of preferred companions shares a group with at least one of them, while minimising surplus familiarity and history-weighted repeated co-membership. Feasibility alone is NP-complete. Because the guarantee is a hard promise, participants can shape it: a set that lists only itself in a cycle is seated as a block. We characterise this *coercion* exactly through closed bipartitions of the preference digraph that the guarantee actually uses in each period, prove with Thomassen’s two-disjoint-cycles theorem that any self-contained set whose submitters list at least three names inside it can be split, show that a minimum-list rule alone does not prevent coercion in stratified periods (odd circulants of step two are coercive, and cross-block padding hides them from any screen run on full lists), and give an exact screen that runs on every period’s effective digraph together with a capacity-demand test and a CP-SAT feasibility certificate. We also show that the guarantee leaks: from sixteen public charts and a known list cap, exact hitting-set inference recovers 29.8% of all directed list entries with no errors. A four-stage solver (anchored construction, repair, closure-move annealing whose accepted moves never increase the number of friendless participants, hinted CP-SAT) meets the guarantee for every participant in every period of a real-scale instance across – independent populations, with –% of participants (mean over seeds) receiving exactly one listed companion and distinct acquaintances – relative to random assignment. The strongest compliant six-person coalition we found, by exhaustive search over its wiring, forces triples but never a table; the stratified attack captures a table in every stratified period until the screen is run on the right graph. Code, traces, experiments and an interactive visualisation are released.

Keywords. group assignment; seating; social mixing; preference elicitation; strategic manipulation; inference from outputs; simulated annealing; constraint programming; digraph partitions.

1 Introduction

Institutions that want their members to know one another rotate them through small groups: schools reassign cafeteria tables every few weeks, universities shuffle discussion sections, companies rotate project teams. The instrument is random assignment, which mixes well but places most members among strangers, and a member who finds no familiar face tends to avoid the group or sit in silence. Attachment theory suggests a remedy short of seating friends together: a single trusted companion who is present, a *secure base*, lowers the anxiety of an unfamiliar setting and frees attention for new contact [4]. The mechanism studied here guarantees one such companion per period and fills the rest of the group with new people.

Each participant may confidentially submit a list of preferred companions. In every period the mechanism partitions the population into groups of prescribed sizes so that every participant who submitted a list shares a group with at least one listed companion (the *anchor guarantee*), keeps listed companions per group to one where it can, and avoids pairs who have already shared a group. Some periods are *stratified*: groups may not mix blocks (grades, departments), which removes cross-block names from the set of usable anchors.

A hard promise can be gamed by shaping the input. Participants who list only each other in a cycle force the mechanism to seat the cycle as a block. We give an exact account of this *coercion*: whether a self-contained set can be broken into two parts each of which keeps every member anchored inside it. Three facts organise the analysis. First, coercion must be judged on the digraph the guarantee actually uses in a period; a set that is harmless on the full lists can be coercive once cross-block names are removed, and we exhibit a rule-compliant six-person attack that captures a table in every stratified period while passing any check run on full lists. Second, a set whose members list at least three names inside it can always be split (Theorem 4.3), so a minimum-list rule of three internal names removes coercion outright, whereas the common rule “at least four names, at least two in your block” leaves internal degree two in stratified periods, exactly where odd circulants of step two are coercive (Proposition 4.5). Third, freedom from coercion is not feasibility: a core of five mutual listers with ten followers passes every structural test yet cannot be seated (Proposition 4.7); a capacity-demand test and a constraint-programming feasibility certificate close that gap.

The guarantee also leaks. Every published chart says of each participant “one of these tablemates is on your list”. Over sixteen periods, with a known cap on list length, exact hitting-set reasoning turns those disjunctions into certainties for a large share of all list entries (Section 5). The earlier claim that charts reveal nothing beyond what participants already know was wrong, and we quantify what they reveal.

Solving the assignment problem is hard even without objectives (Theorem 3.2). We give a hybrid method whose local-search phase moves *closed groups* of participants and rejects any move that would increase the number of participants without an anchor, so that once a feasible assignment is reached the search never leaves the feasible region, and whose final phase is an exact constraint-programming model seeded with the local-search solution as a complete hint. Everything is deterministic in a seed on pinned software versions.

1.1 Contributions

- (i) A model of rotating group assignment with an anchor guarantee, block stratification and a history-weighted objective, and NP-completeness of feasibility (Section 3).
- (ii) An exact theory of coercion on effective digraphs: forcing (Proposition 4.2), splittability from internal degree three via Thomassen’s theorem (Theorem 4.3), coerciveness of odd step-two circulants (Proposition 4.5), a stratified capture attack that complies with the “four names, two same-block” rule, and a capacity-demand condition (Proposition 4.7); a screen that is exact on self-contained candidates of every period’s effective digraph, plus a feasibility certificate (Section 4).
- (iii) A quantified leakage analysis of the published charts (Section 5).
- (iv) A solver whose local-search stages provably never increase the number of friendless participants (Proposition 6.1), with a hinted CP-SAT polish and explicit handling of infeasible inputs (Section 6).
- (v) A reproducible evaluation on a real-scale instance: five scenarios, – independent populations, budget and community sensitivity, structural attacker bounds obtained by exhaustive search, and outcome tails by popularity (Sections 7–8).

2 Related work

Seating and table assignment have been studied as optimisation problems: Lewis and Carroll [10] model wedding and gala seating with likes and dislikes and compare a heuristic with integer programming; Ceylan et al. [5] map the complexity of seat arrangement under various preference structures. Our setting adds repetition over periods with a history-dependent objective, a per-participant hard guarantee instead of a sum of pairwise utilities, and strategic participants. Repeated grouping with a mixing objective is the social golfer problem, a benchmark for symmetry breaking in constraint programming [6]; there repeats are forbidden, here they are penalised because sixteen periods with groups of six or seven in a population of 257 make some repetition unavoidable. Our local-search stage follows simulated annealing [9, 11] and the exact stage uses CP-SAT [12].

Hardness of feasibility is inherited from graph factor problems [7, 8]. The coercion analysis rests on partitioning digraphs under out-degree constraints: a closed part in our sense is a set inducing a subdigraph of minimum out-degree at least one, a closed bipartition is a $(1, 1)$ -splitting in the terminology of Alon [1], and the tool that makes the analysis work is Thomassen’s theorem that minimum out-degree three forces two vertex-disjoint cycles [15]. The complexity of related 2-partition problems is studied by Bang-Jensen and Havet [2], Bang-Jensen et al. [3]; the sets we must decide are small enough for exact enumeration or a tiny SAT model. Closed sets are dual to kernels of digraphs [13]; undirected degree-constrained decompositions go back to Stiebitz [14]. The leakage analysis is a hitting-set argument and uses nothing beyond the published constraints.

3 The rotating group assignment problem

3.1 Instances

A population V of N participants is partitioned into *blocks* by $g : V \rightarrow \mathcal{B}$. There are R periods. Period r specifies groups $\mathcal{T}_r$ with capacities $c_t^{(r)}$, $\sum_t c_t^{(r)} = N$, and is either *unstratified* (any participant may join any group) or *stratified* (group t carries a block label $\beta_t^{(r)}$ and holds only that block, with $\sum_{t:\beta_t=b} c_t^{(r)} = |g^{-1}(b)|$). Each participant i submits a list $L_i \subseteq V \setminus \{i\}$, possibly empty; i is a *submitter* if $L_i \neq \emptyset$. The *effective* list in period r is $L_i^{(r)} = L_i$ in an unstratified period and $\{j \in L_i : g(j) = g(i)\}$ in a stratified one; D_r denotes the *effective digraph* with an arc $i \rightarrow j$ whenever $j \in L_i^{(r)}$.

3.2 Assignments, anchors and history

An assignment for period r is $\sigma_r : V \rightarrow \mathcal{T}_r$ with $|\sigma_r^{-1}(t)| = c_t^{(r)}$ and, in stratified periods, $g(i) = \beta_{\sigma_r(i)}^{(r)}$. Let $s_i^{(r)} = |\{j \in L_i^{(r)} : \sigma_r(j) = \sigma_r(i)\}|$. The *anchor guarantee* is

$$s_i^{(r)} \geq 1 \quad \text{for every submitter } i \text{ and every period } r. \quad (1)$$

A submitter whose effective list is empty makes the period infeasible; the rule “at least two same-block names” exists to prevent that, and the solver reports such an input rather than dropping the participant. History accumulates across periods: m_{ab} counts earlier periods in which a and b shared a group while neither listed the other, and α_{ab} counts earlier periods in which they shared a group while one of them listed the other. (Counting both at the time of seating, rather than reclassifying old meetings by today’s lists, keeps the history meaningful if lists change.)

3.3 Objective

Among assignments satisfying (1) the mechanism prefers one familiar companion and new faces otherwise, and prefers not to repeat co-memberships, with incidental repeats charged more than repeats between listed companions, whose co-seating the guarantee itself requires. We maximise

$$E(\sigma_r) = \sum_i \left(M \mathbf{1}[s_i^{(r)} \geq 1] - \max(0, s_i^{(r)} - 1) \right) - \lambda \sum_{\{a,b\}: \sigma_r(a)=\sigma_r(b)} (\kappa_m m_{ab} + \kappa_\alpha \alpha_{ab}), \quad (2)$$

with M large enough to dominate the other terms and $\lambda, \kappa_m, \kappa_\alpha > 0$; the case study uses $M = 1000$, $\lambda = 1/10$, $\kappa_m = 5$, $\kappa_\alpha = 2$, so an incidental repeat costs 0.5 per prior meeting and a listed one 0.2. Assignments are produced greedily period by period; the history terms couple the periods.

Problem 3.1 (Rotating group assignment with an anchor guarantee). Given an instance, produce $\sigma_1, \dots, \sigma_R$, each satisfying (1) and maximising (2) given the history of the earlier periods.

3.4 Complexity

Theorem 3.2. *Deciding whether an assignment satisfying (1) exists is NP-complete, already for one unstratified period, mutual lists and all capacities equal to three.*

Proof. Membership in NP is clear. For hardness let H be an undirected graph on $3q$ vertices, $V = V(H)$, $L_i = N_H(i)$, and take q groups of capacity three. An assignment satisfies (1) exactly when every group induces a subgraph of H without isolated vertices, that is, a connected three-vertex subgraph, which is a path P_3 or a triangle, and a triangle contains a spanning P_3 . So a feasible assignment exists iff H has a P_3 -factor, which is NP-complete to decide [8]. $\square$

4 Coercion, capacity and screening

4.1 Closed sets and forcing

Fix a period and its effective digraph $D = D_r$.

Definition 4.1. $A \subseteq V$ is *closed* if every submitter in A has an out-neighbour in A ; participants with an empty effective list impose nothing and never break closedness. $S \subseteq V$ is *self-contained* if $L_i^{(r)} \subseteq S$ for every $i \in S$. A self-contained S with at least two submitters is *coercive* if there is no partition $S = A \uplus B$ into closed parts that both contain a submitter.

The *closure* $\text{cl}(i)$ of a participant is the set reachable from i along arcs of D , including i ; closures are self-contained, and every self-contained set is a union of closures.

Proposition 4.2 (forcing). *If S is self-contained and coercive then every assignment satisfying (1) places all submitters of S in one group.*

Proof. Suppose a feasible σ separates the submitters of S . Let x be one of them, $A = S \cap \sigma^{-1}(\sigma(x))$ and $B = S \setminus A$; both contain submitters. A submitter in A has an anchor in its group, and by self-containment that anchor lies in S , hence in A . A submitter in B has an anchor in its own group, which is not x 's group, and that anchor lies in S , hence in B . So A and B are closed, contradicting coerciveness. $\square$

If the submitters of a coercive set fit one group, the mechanism seats them as a block in every period in which the set is self-contained; if they do not fit, the period is infeasible. The one-name cycle of Section 1 is the smallest example. Coerciveness is a property of the effective digraph: a set that is not self-contained on the full lists can become so when cross-block names are removed.

4.2 Splitting

Theorem 4.3 (splitting). *Let S be self-contained, let S' be its submitters, and suppose every member of S' has at least three out-neighbours in S' . Then S admits a partition into two closed parts that both contain a submitter; in particular S is not coercive.*

Proof. $D[S']$ has minimum out-degree at least three, so by Thomassen's theorem [15] it contains two vertex-disjoint directed cycles C_1, C_2 . A cycle is closed (each vertex has its successor). Start with $A = C_1$, $B = C_2$, and repeatedly move any unassigned member of S' that has an out-neighbour in A or in B into that part; both parts stay closed. When no such member remains, every unassigned member v has all its out-neighbours outside $A \cup B$, hence inside the unassigned set Rm , so Rm is closed; add it to A . Finally distribute the non-submitters of S arbitrarily; they do not affect closedness. Both parts contain the submitters of a cycle. $\square$

Remark 4.4. An elementary argument gives the same conclusion for small sets without Thomassen's theorem: if each of the n submitters lists at least $k \geq 3$ names inside S and $n \leq 2k$ then $kn > \binom{n}{2}$ forces a mutual pair $\{x, y\}$, which is closed, and every other submitter keeps at least $k - 2 \geq 1$ names outside the pair. The theorem removes the size restriction, so a rule requiring three names *inside the effective digraph* would make coercion impossible at every size.

The rule used by the school, and by many similar mechanisms, is weaker: at least four names in total, at least two in the participant's own block. In unstratified periods every submitter therefore has out-degree at least four and Theorem 4.3 applies to every self-contained set; in stratified periods the effective out-degree can be two, and degree two is exactly where coercion reappears.

Proposition 4.5 (odd circulants). *Let $C_g(1, 2)$ be the digraph on $\{0, \dots, g-1\}$ in which i lists $i+1$ and $i+2 \pmod{g}$. For odd $g \geq 5$ it is coercive; for even g it is not.*

Proof. A finite closed set contains a directed cycle. In $C_g(1, 2)$ every arc advances the index by one or two around the circle, so a cycle advances by a positive multiple of g in at most two units per step and has length at least $\lceil g/2 \rceil$. Two disjoint closed parts would contain two disjoint cycles with at least $2\lceil g/2 \rceil = g+1$ vertices when g is odd. For even g the parts of even and odd indices are both closed. $\square$

Example 4.6 (stratified capture). Let five juniors $c_0, \dots, c_4$ list $c_{i+1}, c_{i+2} \pmod{5}$ and two distinct seniors each, and let a sixth junior list $c_0, \dots, c_3$. Every list has four names and at least two same-block names. On the full lists the six-set is not self-contained (it names ten seniors whose own closures are large), and any screen run on full lists sees nothing. In a stratified period the seniors vanish from the effective lists, $\{c_0, \dots, c_4\}$ is self-contained and is $C_5(1, 2)$, hence coercive by Proposition 4.5; by Proposition 4.2 the five share a group, and the sixth junior, whose every effective name is among them, is seated with them. All six are forced onto one table in every stratified period (Section 8 confirms this on the full instance). A seven-junior version with two distinct seniors each is also coercive and needs no sixth member.

4.3 Coercion is not feasibility

A set can be splittable in the sense above and still make the instance infeasible, because splitting ignores capacities and the demand that followers place on a core.

Proposition 4.7 (demand). *Let C be self-contained with n_1 submitters and n_0 non-submitters, and let $F(C)$ be the set of submitters outside C whose entire effective list lies in C . In any feasible assignment the members of $C \cup F(C)$ occupy at most $n_0 + \lfloor n_1/2 \rfloor$ groups. Hence if $|C| + |F(C)|$ exceeds the total capacity of that many largest groups admissible to them, no feasible assignment exists.*

Proof. A group containing a follower contains an anchor of the follower, which lies in C . A group containing a submitter of C contains an anchor of that submitter, which lies in C too, so such a group holds at least two members of C . Every group meeting $C \cup F(C)$ therefore contains a member of C , and contains two of them unless its only member of C is a non-submitter. The bound on the number of groups follows. $\square$

Example 4.8. Five participants who list each other (a K_5) and ten followers who each list four of the five pass every structural test (the K_5 splits into a pair and a triple) but need fifteen seats at groups that each hold at least two of the five, so at most two groups, with at most fourteen seats. CP-SAT proves the 257-participant instance infeasible in a few seconds.

4.4 The screen

The screen runs once on the submitted lists, before the first period, and separately on the effective digraph of every period state that will occur (here: unstratified, and stratified with cross-block names removed).

Candidates. The candidates are the closures $\text{cl}(i)$ of all submitters, with size at most 64 (closures larger than that are the ordinary population; a coercive one would exceed every capacity and is reported by the feasibility certificate below). Every candidate is self-contained.

Coercion test. A candidate S is flagged exactly when it is coercive. For $|S| \leq 20$ the test enumerates the $2^{|S|-1}$ subsets containing a fixed submitter and checks both parts for closedness on bitmasks, stopping at the first valid split. For larger S , if every submitter has at least three out-neighbours among the submitters of S the candidate passes by Theorem 4.3; otherwise a small CP-SAT model decides the existence of a closed bipartition exactly (one Boolean per member, one clause per submitter). A candidate with a single submitter is never coercive.

Demand test. For every candidate C the bound of Proposition 4.7 is evaluated against the capacities admissible in that state; a violation is flagged with the core and its followers.

Outcome. Submitters of flagged coercive sets and the submitters and followers of over-demanded cores are *returned* for diversification; violations of the minimum-list and same-block rules and submitters with an empty effective list are reported alongside. Passing the screen is not a feasibility certificate. Before each year the hard constraints alone are solved by CP-SAT for both states with a deterministic budget, and the status (OPTIMAL, INFEASIBLE or UNKNOWN) is recorded; an infeasible instance stops the pipeline with the certificate rather than an assignment.

What the screen does and does not establish. On self-contained candidates the coercion verdict is exact, and by Proposition 4.2 a flagged set is one every feasible assignment would seat as a block. Whether every set that some mechanism is forced to seat together is contained in a self-contained candidate of size at most 64 is not established; a forced set whose members list outsiders with large closures would escape. We know no such example and no proof that none exists. The screen's unit is the effective digraph of each state, which is what closes Example 4.6; the test would also catch an attack hidden in any other systematic filtering of names.

Properties. Tests verify on wirings written in closed form that complete digraphs on 4–12 vertices never flag, one-name cycles and hub wirings (everyone lists a hub and the next member of a chain) always flag, $C_g(1, k)$ with $k \geq 3$ never flags, $C_g(1, 2)$ flags exactly for odd g , a set of six listing each other three times plus one shared outsider is not flagged, six members who list the next member and one shared non-submitting outsider are flagged (the outsider cannot

Algorithm 1 One period

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1: reject the input if a submitter has an empty effective list (InfeasibleInputError)
2:  $\sigma \leftarrow \text{ANCHOREDCONSTRUCTION}$  ▷ Stage 1
3:  $\text{REPAIR}(\sigma)$  ▷ Stage 2: violated set may only shrink
4:  $\sigma \leftarrow \text{CLOSUREANNEAL}(\sigma, I)$  ▷ Stage 3: violation count never increases
5: for up to 6 attempts while some submitter has  $s_i = 0$  do
6:    $\text{REPAIR}(\sigma); \sigma \leftarrow \text{CLOSUREANNEAL}(\sigma, I/2)$ 
7: end for
8:  $\sigma' \leftarrow \text{CPSAT}(\text{hint} = \sigma)$  ▷ Stage 4, previous-period pairs
9: if  $\sigma'$  exists, satisfies (1), and  $C(\sigma') \leq C(\sigma)$  then  $\sigma \leftarrow \sigma'$ 
10: end if
11: if some submitter still has  $s_i = 0$  then
12:    $\sigma' \leftarrow \text{CPSAT}(\text{hint} = \sigma)$  with six times the budget; keep it only if it satisfies (1)
13: end if
14: validate  $\sigma$ ; raise if any submitter is friendless ▷ nothing invalid is ever returned
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separate them), a lone submitter with four non-submitting names is not flagged, a reciprocal pair with three outside names each is not flagged, and the stratified attack of Example 4.6 is flagged on the stratified digraph and passes on the unstratified one.

5 Leakage from the charts

Lists are confidential, but the charts are not, and the guarantee is public knowledge. For participant i and period r let $T_{i,r}$ be i 's group-mates. Anyone who knows that i submitted a list learns the disjunction $\sum_{j \in T_{i,r}} \ell_{ij} \geq 1$, where ℓ_{ij} indicates that j is on i 's list. If lists are fixed for the year and hold at most K names, i 's list is a hitting set of size at most K for the family $\{T_{i,r}\}_r$. A name j is *forced* if every such hitting set contains it, which is decided exactly by computing the minimum hitting set of the family with j removed (a bitmask dynamic programme over the R periods) and comparing with K .

We run this inference on every exported trace and score it against the true lists, which the inference never reads. On the committed honest year (7 as seed, $R = 16$, $K = 8$) it forces 613 directed list entries, 29.8% of the 2056 submitted, all 613 of them correct, touching 243 of 257 participants; 0 participant's entire list is determined. Across – independent populations the count ranges from – to –. The leakage is inherent: any mechanism that publishes charts satisfying a hard guarantee publishes the disjunctions, and repeated periods sharpen them. What can be varied is the side information: an unannounced or larger cap on list length weakens the exact deductions (they rely on K), lists that may change between periods break the fixed-list assumption, and the mechanism should never publish which tablemate served as the anchor. Participants should be told that a determined observer of a year of charts can infer a substantial fraction of their list; the earlier assurance that charts reveal only what participants already know does not hold.

6 Solution method

Each period is solved by the four-stage pipeline of Algorithm 1. Stages 1–3 keep a complete assignment and minimise $C = -E + \text{const}$ with the full history; Stage 4 is an exact model over the previous period's pairs whose solution is kept only if it does not worsen the full-history cost. Every term of C is scaled by ten to stay integral and the temperature is scaled identically, so acceptance probabilities are unchanged.

6.1 Group costs, moves and the monotonicity invariant

For a group with member set T ,

$$\begin{aligned} \text{cost}(T) &= \sum_{i \in T} \text{pen}_i(s_i) + \sum_{\{a,b\} \subseteq T} W_{ab}, & W_{ab} &= \lambda(\kappa_m m_{ab} + \kappa_\alpha \alpha_{ab}), \\ \text{pen}_i(s) &= \begin{cases} M & s = 0 \text{ and } i \text{ is a submitter} \\ \max(0, s - 1) & \text{otherwise,} \end{cases} \end{aligned}$$

so $C(\sigma) = \sum_t \text{cost}(\sigma^{-1}(t))$. The state also caches, per group, the number of friendless submitters, and $\nu(\sigma)$ denotes their total. Every move exchanges a set G_a in group t_a with an equally sized set G_b in group t_b of the same block class; its effect on C and on ν is computed exactly by re-costing the two groups (at most seven members each).

Proposition 6.1 (monotonicity). *Stage 3 never accepts a move with $\Delta\nu > 0$. Consequently, once $\nu(\sigma) = 0$, every assignment visited by Stage 3 satisfies (1).*

Proof. Every candidate move, of any type, is discarded before the Metropolis test if the recomputed friendless count of the two affected groups exceeds the cached one; groups not involved in the move are unchanged. From $\nu = 0$ only moves with $\Delta\nu \leq 0$ are accepted, so ν stays 0. $\square$

This is the invariant we claim for the local search, and it is checked by a test that instruments the accept step. It is deliberately weaker than “moves preserve every individual’s guarantee”: while $\nu > 0$ a move may transfer a violation from one participant to another, which is how a chain is assembled. Closure moves (below) make feasible proposals frequent; the filter makes the invariant hold regardless of how a group was built.

6.2 Stage 1: anchored construction

Participants are processed in increasing order of effective list size, so the participant with the fewest options is seeded first, $i^* = \arg \min_{i \in U} |L_i \cap U|$ over the unmatched set U . Each i is paired with an unmatched listed companion j , preferring one who lists i back, then the smallest W_{ij} , then the shortest list; a companion who does not list i back and has strictly fewer options than i is never taken. Unmatched participants become singleton pods. Pods are then merged until every submitter has a listed companion inside their own pod, needy participants first in increasing order of list size, each merging into the pod of a listed companion if the merged size does not exceed the smallest capacity, choosing by the number of extra familiar faces created, then accumulated history, then size; this assembles one-directional chains before anyone else claims their seats. Pods are placed largest first into the group with enough free seats (and the right block) that minimises accumulated history plus one unit per list relation with the occupants, ties toward emptier groups; a pod that fits nowhere is split into singletons. The result fills every group exactly.

6.3 Stage 2: repair

For up to twenty rounds, each friendless submitter i evaluates the single swap with every non-anchor member of every group holding one of i ’s listed companions, and applies the cheapest swap after which i has an anchor *and* the set of friendless submitters is a strict subset of what it was; a swap that satisfies i by stranding someone else is never accepted. If no such swap exists, the compound move $\text{KICKANDREPAIR}(i, j)$ is tried for i ’s companions j : the closure of i is forced into j ’s group, an equally sized set of j ’s group-mates outside j ’s closure is evicted to i ’s former group, every evicted or stranded participant is re-seated by its own cheapest non-stranding swap, and the compound is kept only if the total cost fell and no participant who was satisfied before is friendless afterwards; otherwise it is undone in place. Repair can stop short of feasibility on

hard instances (a twelve-participant instance from the audit of an earlier draft has a feasible seating that no single swap reaches); Stage 3 and Stage 4 then take over.

6.4 Closures

Moving one participant away from their only anchor strands both, so the annealer moves *closures*. The closure of a set G within its group T is the smallest superset of G stable under two rules: any submitter of $T \setminus G$ all of whose anchors in T lie in G joins G ; any submitter in G without a listed companion inside G pulls in one of its anchors in T (chosen at random). Closures larger than the largest capacity are abandoned. Moving a closure orphans no one who stays and keeps every mover with an internal companion whenever it has one in T . When two closures of different sizes are to be exchanged, the smaller is padded with a random group-mate *and re-closed*, so padding cannot strand the participants the padded member anchored; if the sizes still differ after four attempts the move is skipped. Padding was not re-closed in an earlier version, which produced infeasible proposals; the monotonicity filter would have rejected them, but the proposals were wasted.

6.5 Stage 3: closure-move simulated annealing

The annealer runs I iterations ($I = 300\,000$ in the case study) of Metropolis acceptance on C , $P(\text{accept } \Delta C > 0) = e^{-\Delta C/T}$, T geometric from 2.5 to 0.02 in energy units, and returns the best assignment visited under the lexicographic order (friendless count, cost). Each iteration draws a participant a and one of four move types: with probability $1/4$ a raw single swap with a uniform b of the same block class; with probability $1/4$ a companion-directed single swap (b is a group-mate of one of a 's listed companions, so participants are often proposed beside a friend); otherwise a closure swap $\text{cl}(a) \leftrightarrow \text{cl}(b)$ with re-closed padding; and, while $\nu > 0$, every eighth iteration a targeted move in which a is a friendless submitter and b sits with one of a 's companions, every sixty-fourth of these being a KICKANDREPAIR. Every move passes the $\Delta\nu \leq 0$ filter before the Metropolis test. An iteration re-costs two groups and takes about $15\,\mu\text{s}$; the stage takes about four seconds.

6.6 Stage 4: exact polishing, certification and failure

The annealed assignment is passed as a complete hint to CP-SAT [12]. With $x_{i,t}$ the assignment indicators over the groups allowed for i and $\ell_i = |L_i^{(r)}|$,

$$\begin{aligned} \sum_t x_{i,t} &= 1 \quad \forall i, & \sum_i x_{i,t} &= c_t \quad \forall t, \\ y_{i,t} &\leq x_{i,t}, & y_{i,t} &\leq \sum_{j \in L_i^{(r)}} x_{j,t}, & \sum_t y_{i,t} &\geq 1 \quad \text{for every submitter } i, \\ \sum_{j \in L_i^{(r)}} x_{j,t} &\leq 1 + (\ell_i - 1) z_i + \ell_i(1 - x_{i,t}) \quad \forall t, & & \text{whenever } \ell_i \geq 2, \end{aligned}$$

so that $z_i = 1$ whenever i sits with two or more listed companions (a per-participant encoding with no pairwise companion variables). For each pair (a, b) that shared a group in the previous period with $W_{ab} > 0$, a repeat indicator obeys $r_{ab} \vee \neg x_{a,t} \vee \neg x_{b,t}$ for every t . The objective is $\min \omega_2 \sum_i z_i + \sum_{(a,b)} \omega_r W_{ab} r_{ab}$ with $\omega_2 = 300$ and $\omega_r W_{ab} = 100(\kappa_m m_{ab} + \kappa_\alpha \alpha_{ab})$. Hints cover every variable. The solver runs in deterministic interleaved mode with a deterministic-time budget and eight workers, which makes the outcome a function of model and hint; with sixteen workers on the eight-core machine used here repeated solves were *not* identical, so worker count is part of the reproducibility contract.

A complete hint does not certify anything: a hint can be complete yet infeasible, and a time-limited solve can return UNKNOWN. The pipeline therefore validates every returned assignment itself. The CP-SAT assignment is accepted only if it satisfies (1) and $C(\sigma') \leq C(\sigma)$ on the full history, which the model cannot see; empirically CP-SAT wins early in a year and the annealer

later (Table 2). If any stage leaves a submitter friendless, CP-SAT is rerun with six times the budget, its answer is again validated, and if the period is still unsolved the pipeline raises an error that names the period; reported solve times include that fallback. On the instances of Section 8 the fallback never ran.

6.7 Baseline and determinism

The comparison baseline is uniformly random assignment respecting the period’s stratification, seeded independently; its closed-form expectation is given in Section 8. All randomness (population, per-period local search, baseline, CP-SAT) is seeded. Re-running the export reproduced the committed traces byte for byte apart from timestamps and timings on the pinned versions (Python 3.14.7, NumPy 2.5.2, OR-Tools 9.15.6755), which are recorded in every trace together with the source commit. Reproducibility across other versions is not claimed.

7 Synthetic population model

Real lists are confidential, so evaluation uses synthetic populations with the structural features that matter for the mechanism: blocks, unequal popularity, partial reciprocity, community structure, and the submission rules’ effect on honest participants. Each participant receives a popularity weight $\pi_i \sim \text{LogNormal}(0, \sigma^2)$ and K true friendships, sampled in random order over participants: when i is processed its open slots are filled with distinct $j \neq i$ drawn without replacement with probability proportional to $\pi_j \cdot (\rho \text{ if } g(j) = g(i) \text{ else } 1)$; each new arc $i \rightarrow j$ is reciprocated with probability p if j has an open slot, so every participant ends with exactly K friends; p is calibrated deterministically to a target reciprocity. Latent groups add community structure: participants of each block are cut into groups of sizes uniform on 6–12 (optionally spanning two blocks); with probability ω a participant joins a secondary group; each friendship draw is assigned to the participant’s group pool with probability μ (popularity-weighted, background draws excluding the own group so the in-group share is μ in expectation). At $\mu = 1$ groups are isolated and small groups yield fewer than K friends. Unit tests confirm the in-group share is within ± 0.05 of μ , the secondary-group frequency within ± 0.05 of ω , and that $\mu = 0$ reproduces the plain block-biased network exactly.

Honest participants submit their true friends subject to the rules. A list with fewer than four names or fewer than two same-block names is *not* a valid submission and is recorded as no submission; the participant keeps no guarantee and the count of such participants is reported rather than hidden by padding the list with acquaintances (padding remains available as an explicit option for modelling participants who name acquaintances to qualify, and is not used here). With $K = 8$ and the 9:1 block bias no participant of the committed cohort is affected (0 rejected, 0 non-submitters); cohorts with shorter or more cross-block friendship lists would be, and that cost belongs in any deployment decision. A planted set of six mutual friends plays the colluding group.

8 Case study: a secondary-school lunch period

8.1 Instance and scenarios

The instance mirrors a real school’s lunch period: $N = 257$ students in two blocks (grade 11: 132, grade 12: 125), 40 groups of which 17 seat 7 and 23 seat 6, $R = 16$ periods alternating unstratified (“mixed”) and stratified (“same-grade”), the stratified partition giving juniors 12 seven-seat and 8 six-seat tables and seniors 5 and 15. Population parameters: $K = 8$, $\sigma = 0.6$, $\rho = 9$, target reciprocity 0.5 (measured 0.490), within-block share 0.904, $\mu = \omega = 0$. Rule: four names, two same-block. Energy parameters as in Section 3; $I = 300\,000$; CP-SAT deterministic-time

Table 1: Headline statistics per scenario (seed 7, 16 rotations). “ ≥ 1 ” and “exactly one” are percentages of submitters; “met” is the mean number of distinct schoolmates a student has shared a table with by rotation 16; cluster statistics describe the six coalition members (mean = $\sum_c |c|^2/6$).

Scenario	≥ 1 (min)	exactly one		met	met, random	intact	mean cluster
		mean	min				
Honest	100.0	99.68	98.4	74.44	72.31	—	—
Coalition, no defenses	100.0	99.59	98.4	72.78	72.31	16/16	6.00
Coalition, min-4 rule (star)	100.0	98.30	97.3	74.00	72.31	0/16	3.02
Coalition, same-grade attack	100.0	98.69	96.9	73.33	72.31	8/16	3.79
Coalition, screens on	100.0	98.30	97.3	74.00	72.31	0/16	3.02

budget 3.5 with 8 workers; feasibility certificates for both states before each year (OPTIMAL / OPTIMAL, 5.0 s together); seed 7. Table 6 collects every parameter.

Five scenarios share the population and differ only in the six colluders’ lists:

- *honest*: everyone submits true friends;
- *no defenses*: the six submit a one-name cycle; no rule, no screen;
- *min-4 rule*: the six submit the strongest 4-of-5 wiring we found, the *omission star* (member c_0 omits c_1 , everyone else omits c_0 , so nobody lists c_0); screen off;
- *same-grade attack*: Example 4.6; rule on, screen off;
- *screens on*: Example 4.6 is screened on both effective digraphs and returned; the group resubmits the strongest wiring that passes (the omission star), which is the adversarial response rather than the honest one, and the year is played with that.

8.2 Attacker strength

For a coalition whose lists stay inside the coalition, every feasible seating induces a partition of the coalition into closed parts, and the member-weighted mean cluster size $\sum_c |c|^2/6$ of the best such partition is a lower bound on the cohesion any solver must grant, independent of seed, budget and history. Table 5 gives these bounds. The one-name cycle admits only the trivial partition (bound 6). The wiring used in an earlier version of this work, whose omitted names form two 3-cycles, can be split into three mutual pairs (bound 2). The omission star cannot: nobody lists c_0 , so c_0 cannot sit in a coalition pair, the part containing c_0 has size at least three, and the only patterns are 3+3, 4+2 and 6, with member-weighted means 3, 10/3 and 6 (bound 3). Exhaustive enumeration of all 5^5 four-of-five wirings (member c_0 ’s omission fixed by symmetry) shows that 3 is the largest bound any such wiring can force (30 wirings attain it). This is the attacker’s best structural guarantee with six members and purely internal lists; it says nothing about adaptive lists, mixed honest and strategic members, larger coalitions, or the use of outsiders as scarce anchors, none of which we model.

8.3 Results

Table 1 summarises the five scenarios on the committed seed; Table 2 follows the honest scenario period by period; Table 3 follows the colluders under each defense level; Table 4 lists the screen’s verdicts; Table ?? reports – independent populations and Table ?? the budget and community sensitivity.

Guarantee, novelty and baseline. The anchor guarantee holds for every submitter in every period of every scenario and every seed (Table ??: —), as asserted before export and re-checked by the tests; this is the checked condition on completed runs, and no run failed or timed out. On the committed seed 99.68% of submitters have exactly one listed companion on average (worst period 98.4%); across seeds the mean is —% (range —). Students meet 74.44 distinct

Table 2: Honest scenario, rotation by rotation. “2+” counts submitters with two or more listed peers at their table; repeat pairs are pairs seated together who had already shared a table this year (proposed / random); “phase” is the pipeline stage whose seating was kept; time is the wall-clock solve time of the rotation including any fallback solve.

rot.	state	≥ 1	exactly one	2+	met	met, rnd	repeats (prop./rnd)	phase	time (s)
1	mixed	100	100.0	0	5.5	5.5	0 / 0	cpsat	6.3
2	same	100	100.0	0	10.9	10.8	4 / 17	cpsat	8.0
3	mixed	100	100.0	0	16.1	16.1	27 / 20	cpsat	8.3
4	same	100	100.0	0	21.2	21.1	56 / 66	anneal	8.7
5	mixed	100	100.0	0	26.2	26.1	59 / 47	anneal	8.4
6	same	100	100.0	0	31.1	30.8	71 / 99	anneal	8.5
7	mixed	100	100.0	0	35.9	35.6	86 / 85	anneal	8.2
8	same	100	99.2	2	40.5	40.0	112 / 135	anneal	8.5
9	mixed	100	100.0	0	45.1	44.7	103 / 108	anneal	8.6
10	same	100	100.0	0	49.6	48.8	134 / 179	anneal	7.9
11	mixed	100	99.6	1	54.0	53.1	124 / 138	anneal	8.3
12	same	100	98.4	4	58.2	56.9	175 / 218	anneal	8.6
13	mixed	100	100.0	0	62.4	61.0	152 / 177	anneal	8.5
14	same	100	98.4	4	66.3	64.8	201 / 214	anneal	8.5
15	mixed	100	100.0	0	70.7	68.8	145 / 180	anneal	9.1
16	same	100	99.2	2	74.4	72.3	217 / 257	anneal	8.7

others by the end of the committed year against 72.31 under the simulated random baseline; the closed-form expectation of the random baseline for this schedule is 72.29 (it would be 74.72 if all sixteen periods were unstratified, so alternation itself costs the baseline about 2.4 acquaintances). Across seeds the difference between the proposed system and its paired random baseline ranges from $-$ to $-$ (mean $-$): the history penalties make the mechanism roughly as good as random assignment at spreading students, not clearly better, and the proposed seating produces more repeat pairs than the random one in 3 of 16 periods on the committed seed (periods 3, 5, 7). The baseline seats 21.2% of submitters with a listed companion in a typical period (closed form 21.8%: 16.0% in unstratified and 27.7% in stratified periods).

Coalition. With no defenses the cycle is seated together in 16/16 periods, as Proposition 4.2 requires. Under the min-4 rule the omission star is never intact (0/16) and its mean cluster is 3.02 (3+3 in 15 rotations, 4+2 in 1 rotations), which matches the structural bound of 3 and shows the solver is not being generous; across seeds the star is intact in $-$ periods in total and its mean cluster ranges from $-$ to $-$. The same-grade attack is scattered in every unstratified period (0/8 intact) and seated as a block in every stratified one (8/8), 8/16 overall, exactly as Example 4.6 predicts, while every list satisfies the rule and a screen on full lists sees nothing. With the screen run on both effective digraphs, the stratified candidates {S015, S030, S055, S110, S115}; {S015, S030, S055, S110, S115, S121} are flagged (Table 4), all 6 colluders are returned and no honest student is; the adversarial resubmission passes and yields mean cluster 3.02 with 0/16 intact.

Leakage and tails. Section 5 gives the leakage numbers. Outcomes are not uniform: on the committed seed the number of distinct schoolmates met ranges from 64 to 82 (quartiles 73.0 and 77.0); the quartile of students least often listed by others (mean in-degree 3.1) meets 75.78 on average and the most-listed quartile (13.9) meets 72.48, so low popularity does not translate into fewer acquaintances under this mechanism, while popular students absorb more repeat demand. These are descriptive facts about one synthetic family, not a fairness result.

Table 3: Coalition outcome per rotation under each defense level: partition pattern of the six members across tables (“6” = all at one table) and the member-weighted mean cluster size.

rot.	state	no defenses		min-4 (star)		same-grade attack		screens on	
		pattern	mean	pattern	mean	pattern	mean	pattern	mean
1	mixed	6	6.00	3+3	3.00	2+1+1+1+1	1.33	3+3	3.00
2	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00
3	mixed	6	6.00	3+3	3.00	2+2+1+1	1.67	3+3	3.00
4	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00
5	mixed	6	6.00	4+2	3.33	2+2+1+1	1.67	4+2	3.33
6	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00
7	mixed	6	6.00	3+3	3.00	2+2+1+1	1.67	3+3	3.00
8	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00
9	mixed	6	6.00	3+3	3.00	2+1+1+1+1	1.33	3+3	3.00
10	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00
11	mixed	6	6.00	3+3	3.00	2+1+1+1+1	1.33	3+3	3.00
12	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00
13	mixed	6	6.00	3+3	3.00	2+1+1+1+1	1.33	3+3	3.00
14	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00
15	mixed	6	6.00	3+3	3.00	3+2+1	2.33	3+3	3.00
16	same	6	6.00	3+3	3.00	6	6.00	3+3	3.00

Table 4: Screen on the coalition’s initial submission (same-grade attack): candidate closed sets on each rotation state’s effective lists and the verdict.

state	candidate closed set S	verdict	closed split found
mixed	(no closed set of at most 64 students)	–	–
same	S015, S030, S055, S110, S115	flagged	none
same	S015, S030, S055, S110, S115, S121	flagged	none

Sensitivity and cost. Table ?? varies the annealing budget by a factor of three either way and the community regime at fixed budget. Solve time per period is 8.8 s on average and at most 10.0 s across the committed scenarios (– s across seeds), on an eight-core desktop; the committed export of five scenarios took 701 s of solver time.

Relation to earlier figures. An earlier proposal for the same school reported about 75% of students with exactly one listed companion and 67–69 distinct acquaintances against 73–75 under random seating. Those figures came from a different mechanism (it rewarded a second companion rather than penalising it), a different schedule (all periods unstratified, whose random baseline is 74.72), and code and data that are no longer available, so they are not comparable with the present ones and are not reproduced here.

9 Discussion

What the rules buy. The minimum-list rule is often described as the defense against private tables. Its actual content is Theorem 4.3: three names *inside the digraph the guarantee uses* make coercion impossible at any coalition size. “Four names, two in your block” delivers that in unstratified periods and not in stratified ones, where two same-block names leave odd circulants coercive; the screen on effective digraphs, not the rule, is what stops Example 4.6. A rule of three same-block names would remove the stratified loophole by theorem, at the cost of excluding honest students with fewer than three same-block friends. Either way, a compliant coalition of six retains a structural cohesion of three: the mechanism gives each of them one familiar face,

Table 5: Structural cohesion bounds for six-member wirings: the minimum, over all partitions of the six into closed parts, of the member-weighted mean cluster size and of the largest cluster. These bounds hold for every feasible seating regardless of solver, seed or history.

wiring	min mean cluster	min largest cluster	passes screen
one name each, cycle ($k=1$)	6.0	6	no (coercive)
4-of-5, omitted names form two 3-cycles	2.0	2	yes
4-of-5, omission star (nobody lists member 1)	3.0	3	yes
best of all 5^5 4-of-5 wirings (exhaustive)	3.0	3	yes

Table 6: Parameters used for the committed traces.

group	parameter	value
cohort	students (grade 11 / 12)	257 (132 / 125)
cohort	true friends per student K	8
cohort	popularity σ (lognormal)	0.6
cohort	within-grade weight	9.0
cohort	reciprocity target / measured / p	0.5 / 0.490 / 0.60
cohort	measured within-grade fraction	0.904
cohort	latent groups μ, ω	0.0, 0.0
rules	minimum list / same-grade names / short lists	4 / 2 / no submission
tables	capacities	$17 \times 7, 23 \times 6$
rotations	count / first state	16 / mixed
stage 3	satisfied submitter	1000.0
stage 3	extra listed peer	1.0
stage 3	repeat per prior meeting: incidental / listed	0.5 / 0.2
stage 3	temperature $T_0 \rightarrow T_1$	$2.5 \rightarrow 0.02$
stage 3	iterations	300,000
stage 4	weight per student with ≥ 2 peers	300
stage 4	repeat scale	100
stage 4	deterministic time budget / workers	3.5 / 8
certificate	feasibility budget (deterministic time)	20.0
run	seed / versions	7 / 3.14.7, 2.5.2, 9.15.6755

then two, never a table.

Guarantees, certificates and promises. A hard guarantee is the right instrument for participants with few options, who are precisely the ones a soft version would fail. Its price is that the mechanism must say clearly what it certifies: a returned assignment satisfies every submitter (validated), a feasibility certificate says whether any assignment can, a screen says which sets would be seated as blocks. None of these is a promise that arbitrary submissions will be seated: the K_5 -plus-followers input is admissible, passes the structural screen, and is infeasible, and a deployment needs an exception process for that case.

Leakage. Confidential collection does not make the output uninformative. The published charts of one year determine a large share of list entries when the list cap is known, and they constrain the rest. A deployment should say so, avoid publishing anchors or the cap, and consider allowing lists to change.

Honest-user costs. The rules exclude honest participants with fewer than four friends or fewer than two in their block; the committed cohort has none, real cohorts will. Returning a flagged group for diversification presumes its members have other acceptable names; an honest

small group and a strategic one can submit identical digraphs, so the screen cannot infer intent, and the exception process has to be human.

Limitations. The population family is synthetic and, although – independent cohorts and two budget levels are reported, no real list statistics calibrate it. The attacker model is static: lists fixed for the year, one coalition of six with internal lists, best-found rather than optimal wirings beyond the exhaustive 4-of-5 search, and an adversarial resubmission chosen by us. The screen’s completeness beyond self-contained candidates is open. Stage 4 sees only the previous period’s pairs. Timing is from one machine and one software version. Nothing here measures belonging or psychological safety in a real pilot.

10 Conclusion

Rotating group assignment with an anchor guarantee is NP-complete to decide, coercible by design, and leaky by design; each of these has a precise form. Coercion is decided exactly on self-contained sets of each period’s effective digraph, is impossible whenever members list three names inside the set, and reappears in stratified periods under the common “two same-block names” rule through odd step-two circulants, which a screen on the right graph catches before the first period. Freedom from coercion is not feasibility, so a demand test and a constraint-programming certificate accompany the screen. The charts leak a quantifiable share of the lists. A hybrid solver whose local search never increases the number of friendless participants meets the guarantee for every participant in every period across independent populations while matching random assignment’s mixing, and the strongest compliant coalition we could construct obtains triples, not a table.

Acknowledgements. An independent audit of an earlier draft (September 2026) produced the stratified capture attack, the K_5 -plus-followers infeasibility, the odd-circulant counterexample, the omission-star wiring, the leakage analysis, and the repair and padding counterexamples; every one of them is addressed above, and the audit’s reproduction scripts informed the regression tests.

Data and code availability. Implementation, committed traces, experiments, tests and the interactive visualisation: <https://github.com/IsaacLin247/lunch-seating-visual>; visualisation at <https://isaacclin247.github.io/lunch-seating-visual/>. Every number in this article is generated from the committed data by a script in the repository.

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A Reproduction

```
python -m venv .venv && . .venv/bin/activate && pip install -r requirements.txt
python sim/export_traces.py # five scenarios, deterministic, ~15 min
python sim/experiments.py --seeds 10 # seeds, budgets, communities -> results/
    experiments.json
pytest # unit tests + trace invariants + audit regressions
python paper/make_tables.py && cd article && latexmk -pdf article.tex
```